

Basic Principles of Medicine 1
Module : Foundations to Medicine
Lecture No: 12

**Title: Mendelian Inheritance: Autosomal
Dominant and Autosomal Recessive (Flipped
classroom)**

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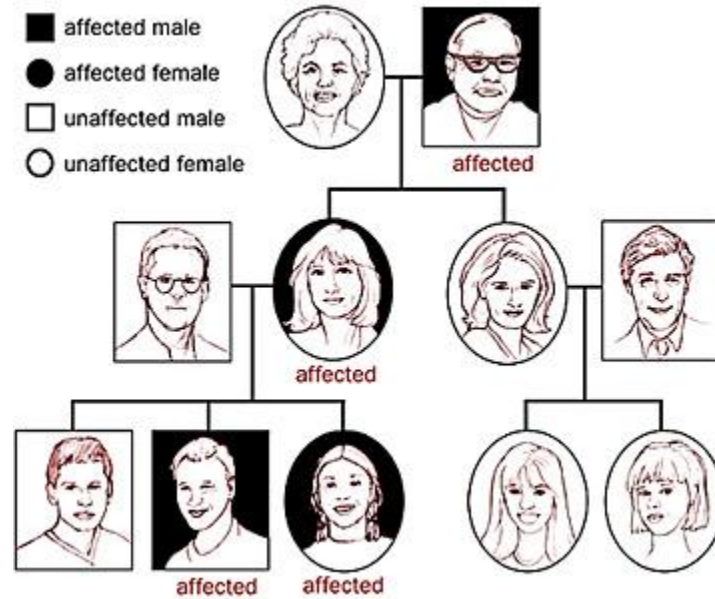
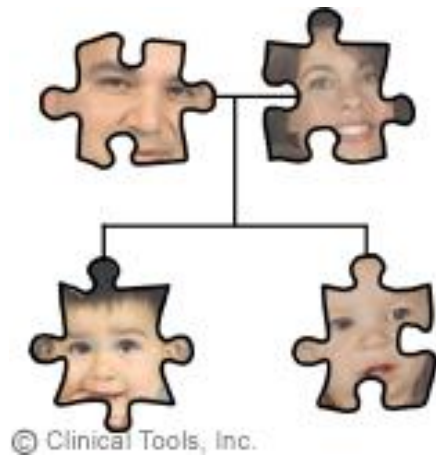
Pre-lecture Required Reading

- **Reading suggestion:** Human Genetics and Genomics: Korf & Irons 4th Edition; Chapter 3 & 10 for Patterns of Inheritance lectures.
 - You can find the above book page in your Sakia site under: Resources/General Information/Book Access
 - Look at self-assessment problems for Chapter 3 (page 63; and answers which are on page 248).
 - Parts I, II, III, and IV of the Neurofibromatosis Chapter 10 of Korf 4th edition (these are in blue boxes). Also, pay attention to Table 10.2 and Figure 10.4
- Review basic genetics concepts
- DLAs linked to this flipped classroom
- Email: shupadhya@sgu.edu (Questions regarding lecture and DLA)

Inheritance: AD and AR – *DLAs*

SOM.MK.III.BPM1.1.FTM.3.GNET.0020	Describe basic definitions of human genetics including: locus, gene, allele, genotype, and zygosity
SOM.MK.III.BPM1.1.FTM.3.GNET.0021	From a description of a pedigree, identify autosomal dominant (AD) and autosomal recessive (AR) inheritance; Compare and contrast Mendelian and non-Mendelian genetics
SOM.MK.III.BPM1.1.FTM.3.GNET.0022	Calculate recurrence risk of autosomal dominant and autosomal recessive disorders within families
SOM.MK.III.BPM1.1.FTM.3.GNET.0023	Analyze family pedigree to determine the risk that immediate family members might be carriers of disease alleles (AR or X-linked) and when to use the 2/3 rule
SOM.MK.III.BPM1.1.FTM.3.GNET.0024	Describe and differentiate between pseudoautosomal dominance, co-dominance, and incomplete dominance; give examples
SOM.MK.III.BPM1.1.FTM.3.GNET.0025	Describe co-dominance, giving examples
SOM.MK.III.BPM1.1.FTM.3.GNET.0026	Describe incomplete dominance, giving examples

Patterns of inheritance: Autosomal dominant and recessive



An important thematic objective of these lectures is to encourage students to learn how to quickly create and interpret a three-generation family pedigree from patient history. From this, many genetic phenomena may be recognized

Inheritance: AD and AR - *Flipped classroom lecture*

SOM.MK.III.BPM1.1.FTM.3.GNET.0027	Interpret a family pedigree incorporating standard genetic symbols. Interpret a pedigree and identify features, disorders, genotype, carrier status, recurrence risk, and genetic principles
SOM.MK.III.BPM1.1.FTM.3.GNET.0028	Demonstrate how disease liability is transmitted for autosomal dominant and autosomal recessive disorders, beginning from the first "de novo" occurrence
SOM.MK.III.BPM1.1.FTM.3.GNET.0029	Propose how the basic risk calculations are performed for the different categories of genetic transmission: AR and AD
SOM.MK.III.BPM1.1.FTM.3.GNET.0030	Identify obligate carriers in a pedigree: AR, or AD with reduced penetrance
SOM.MK.III.BPM1.1.FTM.3.GNET.0031	Distinguish the mode of inheritance, identify the main clinical features, and interpret genetic principles illustrated by the following autosomal dominant disorders: Huntington disease, myotonic dystrophy, familial hypercholesterolemia, osteogenesis imperfecta, Marfan syndrome, Achondroplasia, NF, acute intermittent porphyria)
SOM.MK.III.BPM1.1.FTM.3.GNET.0032	Distinguish the mode of inheritance, identify the main clinical features, and interpret genetic principles illustrated by the following following autosomal recessive disorders: CF, PKU, Tay-Sachs disease, Sickle cell disease, thalassemia, hemochromatosis, homocystineuria, SCID due to ADA deficiency, alkaptonuria
SOM.MK.III.BPM1.1.FTM.3.GNET.0033	Identify pseudodominant inheritance and indicate factors that result in this inheritance pattern. Be able to identify this concept from clinical scenarios and pedigrees
SOM.MK.III.BPM1.1.FTM.3.GNET.0034	Compare and contrast locus heterogeneity and allelic heterogeneity; give examples
SOM.MK.III.BPM1.1.FTM.3.GNET.0035	Explain how allelic heterogeneity allows the formation of a compound heterozygote in autosomal recessive disorders
SOM.MK.III.BPM1.1.FTM.3.GNET.0036	Define the terms and give examples for: haploinsufficiency, dominant negative effect, gain of function mutations, loss of function mutations

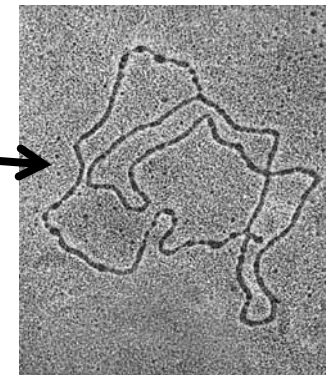
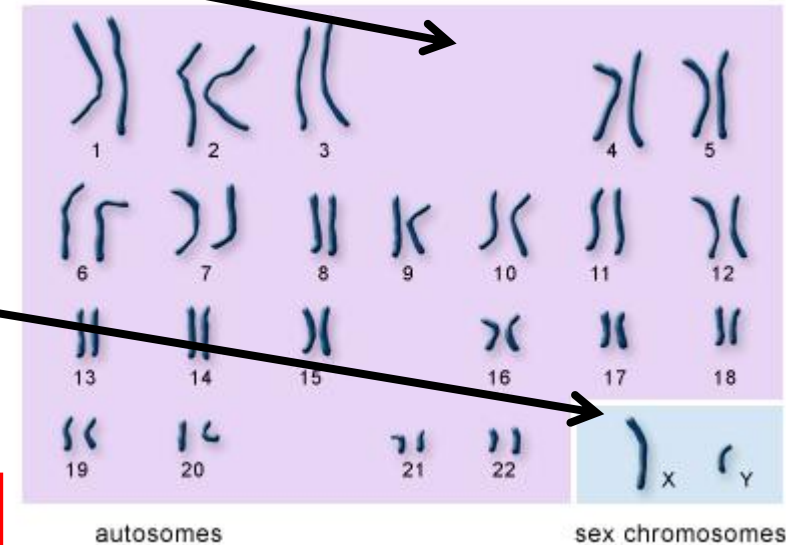
Patterns of Inheritance: Consists of 3 lectures (most set up as flipped classroom), and 2 blocks of DLA material; many clickers presented in class, and practice questions

Modes of Inheritance

MENDelian

- Autosomal Dominant
- Autosomal Recessive
- X – Linked dominant
- X – Linked recessive
- Y – linked inheritance
- Mitochondrial (due to a mutation in a gene encoded by mitochondrial DNA)

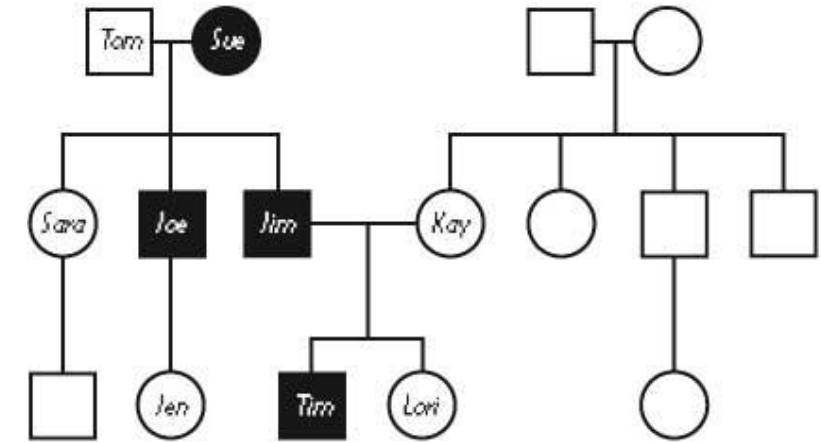
Non-Mendelian



A pedigree from a family with neurofibromatosis is shown. Predict the genotypes of the individual II-2 and II-3. (Assume A is the normal allele and 'A*' is the mutant allele)

Additional Q: What is the risk of the next child of II-3 and II-4 being affected?

- A. AA, A*A*
- B. AA*, AA*
- C. A*A*, A*A*
- D. AA, AA
- E. A*A*, AA*
- F. AA*, AA



17%
17%
17%
17%
17%
17%



Autosomal Dominant disorders

Slide deck from DLA has details for each disorder

- Familial hypercholesterolemia
- Huntington disease
- Myotonic dystrophy
- Marfan syndrome
- Osteogenesis imperfecta
- Achondroplasia
- Neurofibromatosis type I
- Acute intermittent porphyria

Gene names when described, inheritance pattern, and diagnosis based on salient clinical features

You will see all of these again as you progress through medical school

DLA video for information about these disorders!

A 35-year-old woman comes to the physician for follow-up examination because she has a family history of early onset heart disease. Physical examination shows painless swellings on the Achilles tendons. Which of the following genes is most likely involved in the disorder in this patient?

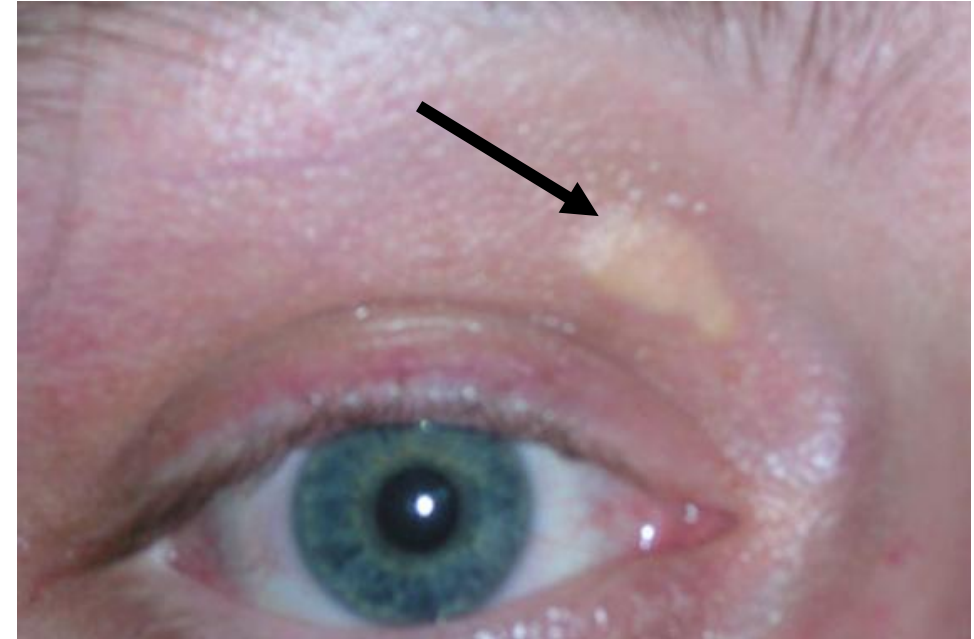
- | | |
|------------------|----|
| A. <i>NF1</i> | 0% |
| B. <i>CFTR</i> | 0% |
| C. <i>LDLR</i> | 0% |
| D. <i>HEXA</i> | 0% |
| E. Huntington | 0% |
| F. <i>COL1A1</i> | 0% |
| G. <i>FGFR3</i> | 0% |

This was discussed in the DLA

What's the risk of her brother having the same pathogenic allele?

Familial hypercholesterolemia (LDL receptor deficiency)

- LDL receptor (encoded by *LDLR*) present on plasma membrane of hepatocytes
- Low density lipoprotein (LDL) binds to the LDLR to allow endocytosis
- LDL is cleared from the blood
- **Haploinsufficiency** for *LDLR* have higher levels of circulating LDL, and therefore **high serum cholesterol**.
- Cholesterol accumulates in
 - Blood vessel wall → Heart disease
 - Xanthomas, xanthelasma



Attribution: Bobtheowl2 at the English Wikipedia, CC BY-SA 3.0,
<https://commons.wikimedia.org/w/index.php?curid=4040642>

Xanthelasma: cholesterol deposits (xanthoma)
that are found near the eye

FHC (*LDLR* deficiency)

- Age of presentation: 30-50 years
- Many different mutation alleles may destroy the function of a gene.
- **Allelic heterogeneity**
 - Different mutations of the **SAME** gene (*LDLR* gene)



Xanthomas: Yellowish deposition of cholesterol near elbows, ankles, wrists, palpebral areas

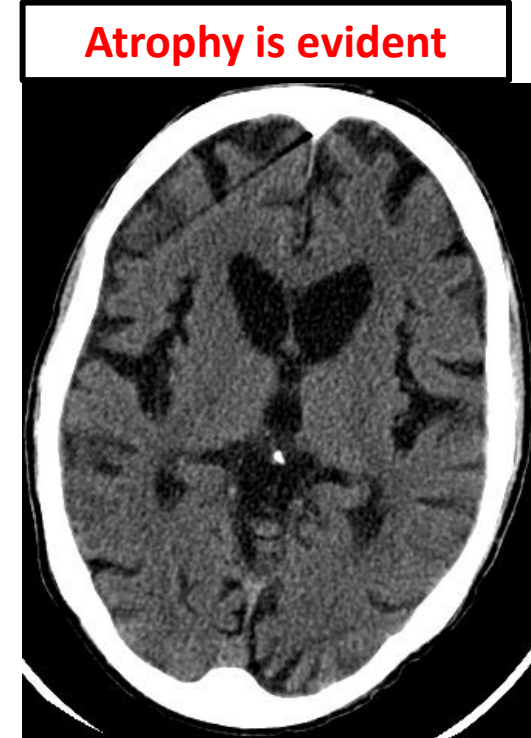
Huntington disease (HD)

- Triplet repeat expansion disorder
 - Neurodegenerative
 - Movement disorder
- “Gain of function” of the protein
 - “Attainment of a novel function”
 - Huntingtin protein accumulation → Leads to death of neurons
- Degeneration of neurons in basal ganglia and cortex of the brain.
- Symptoms: Chorea, cognitive decline, memory loss, sleep disorder...
- Age of presentation: 30-50 years

CT Scan



Person without HD



Person with HD

Chorea: jerky involuntary movements

Myotonic dystrophy

- Triplet repeat expansion in the *DMPK* gene
- An example of a **triplet repeat expansion disorder** that has a **pleiotropic** phenotype
- Pleiotropy: Many different features (phenotypic manifestations) affecting multiple organs, all ascribed to a single genetic cause
- Autosomal dominant transmission
- Characterized by wasting of the muscles, cataracts, heart conduction defects, endocrine changes, and myotonia.



*Myotonia is reduced ability to relax after a muscle contraction.
Don't confuse myotonia with myoclonus which is a quick, involuntary muscle jerk.*

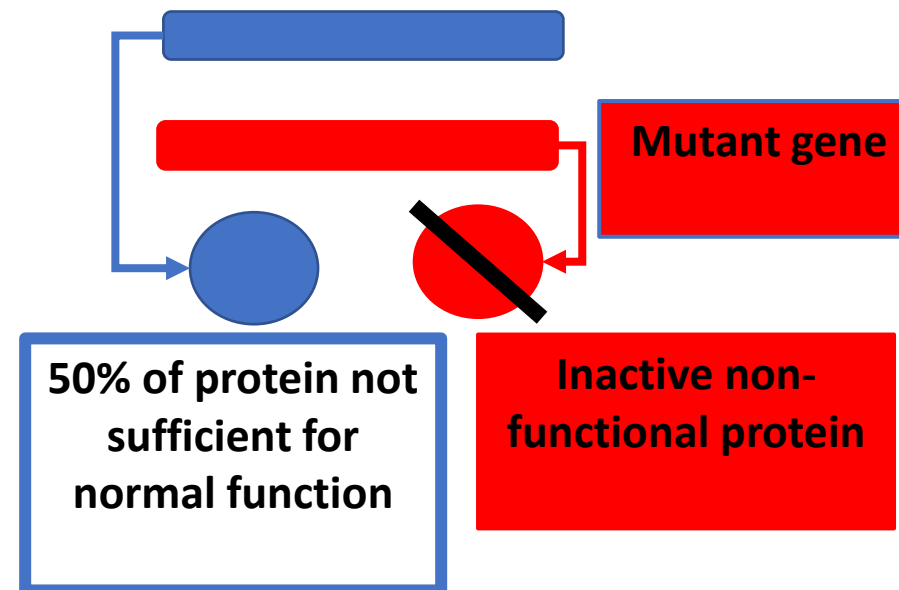
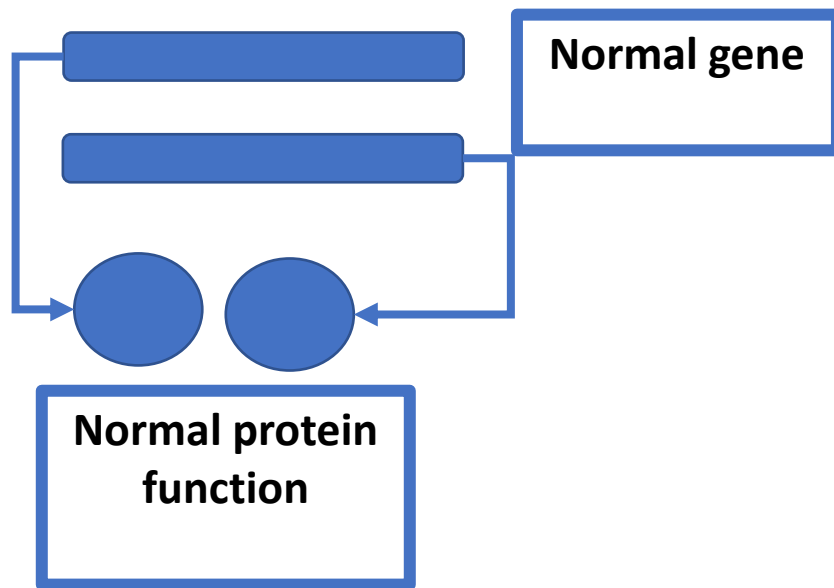
A 28-year-old man comes to the physician for follow-up examination because he has a family history of early myocardial infarction. His grandfather and mother have elevated serum cholesterol levels. Physical examination shows bilateral eyelid xanthelasma. Genetic testing shows a pathogenic mutation of one copy of the *LDLR* gene. Which of the following best explains the inheritance of the disorder in this patient?

- | | |
|-----------------------------|-----|
| A. Triplet repeat expansion | 14% |
| B. Autosomal recessive | 14% |
| C. Pseudodominance | 14% |
| D. Gain of function | 14% |
| E. Haploinsufficiency | 14% |
| F. Dominant negative | 14% |
| G. Variable expressivity | 14% |

This was discussed in the DLA

Molecular explanation: why AD mutations manifest in the heterozygous state

- **Haplo-insufficiency:** Loss-of-function mutations in which half normal levels (50%) of the gene product result in phenotypic effects. Reduced protein levels (50%) are not sufficient to carry out the normal functions of that protein
 - Examples include some cell membrane receptors (familial hypercholesterolemia)
 - Acute intermittent porphyria (Rare example of an autosomal dominant enzyme deficiency); (heme can't be produced fast enough; and pathway intermediates build up and cause disease)
 - Osteogenesis imperfecta (OI) type I (half the amount of collagen causes brittle bones. OI type I discussed in more detail in the FTM-2 and MSK module)



Autosomal dominance and haploinsufficiency of an enzyme (Rare example)

- Acute intermittent porphyria (AIP) (*HMBS* gene*)
- Must have two functional alleles of *HMBS* because the enzyme synthesizes the porphyrin ring and heme
 - A lot of heme is needed in cells
- Most inborn errors of metabolism (due to enzyme deficiency would be recessive: either AR, or X-linked)
- A very small amount of enzyme activity is sufficient to meet needs of cells
 - Note 50% activity of enzyme (in autosomal recessive disorders) → No clinical features of the disorder → Carriers of the mutation

****At this point, there is no need to memorize HMBS as the gene name.
HMBS will be described in more detail in the BPM2 DM module, during description of
the heme biosynthesis pathway***

Marfan syndrome- Autosomal dominant

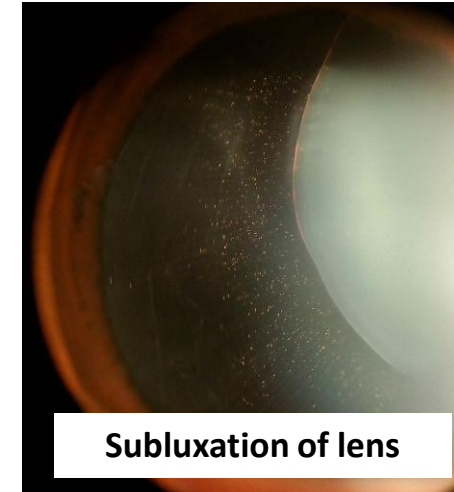
- Fibrillin gene (FBN1 gene) mutations
- Impaired connective tissue assembly
- Pleiotropic phenotype is related to connective tissue:
 - Chest wall deformity
 - Tall stature, arachnodactyly
 - Risk of heart defect
 - Eye lens subluxation

Read about Marfan syndrome on page 45-46 (Korf 4th edition)

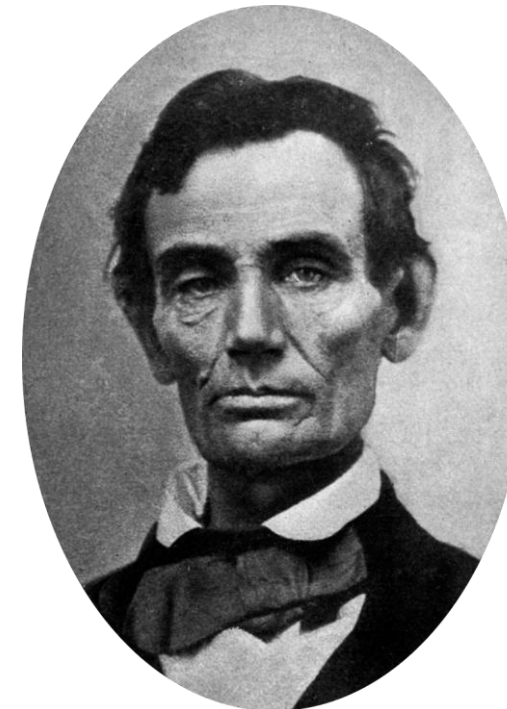


Pectus excavatum

By Aurora Bakalli, Tefik Bekteshi, Merita Basha, Afrim Gashi, Afërdita Bakalli and Petrit Ademaj - <http://casesjournal.biomedcentral.com/articles/10.4076/1757-1626-2-8827>, CC BY-SA 3.0, <https://commons.wikimedia.org/w/index.php?curid=50545526>

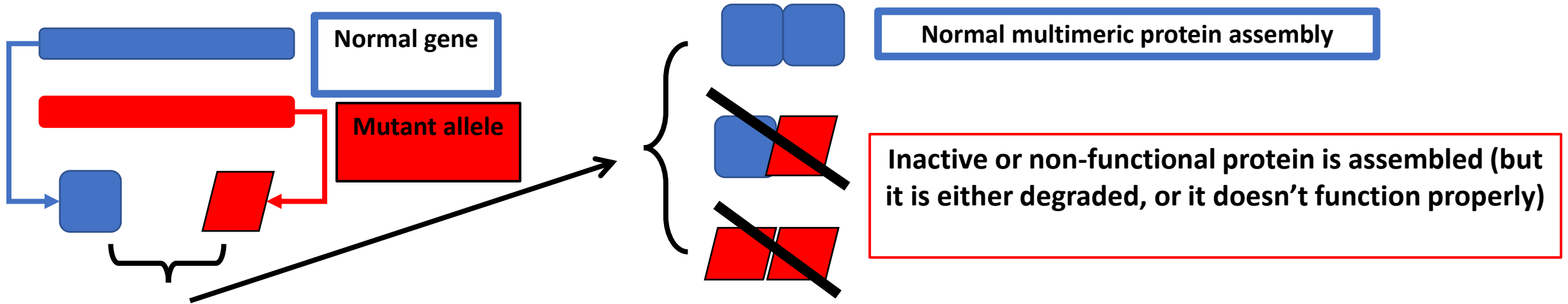


Subluxation of lens



T.P. Pearson, Macomb, IL [Public domain],
via Wikimedia Commons

Molecular explanation: why AD mutations manifest in the heterozygous state



- **Dominant-negative mutations:** A mutant gene product interferes with function of the normal gene product; In some cases, assembly of the **multimeric protein** is affected (hindered) by the presence of the mutant protein
 - Examples: Collagenopathies such as severe OI Type II, III, or IV; also, Marfan syndrome (FBN1 mutation)
 - Sometimes dominant negative mutations described as “special case” of gain of function
 - Typically, dominant negative mutations produce more severe phenotype than haploinsufficiency
 - **Marfan syndrome and OI: Caused by either haploinsufficiency, or dominant negative type mutations**

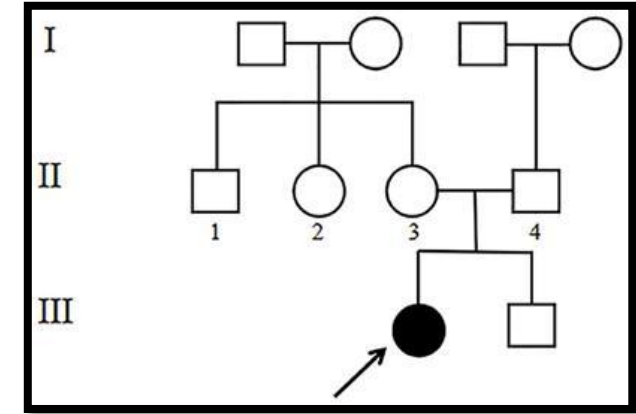
Marfan syndrome- Autosomal dominant

Severe forms of Marfan syndrome explained by → **dominant negative**

- Mutation in *FBN1* gene (fibrillin-1) causes defect in connective tissue.
- For dominant negative, both alleles are expressed (autosomal)
 - The protein formed from mutant *FBN1* allele doesn't assemble properly with other proteins
 - Interferes with higher order protein structure formation in connective tissue
 - More severe phenotype (less than 50% functional fibrillin-1)

Read about Marfan syndrome on page 45-46 (Korf 4th edition)

Genetic testing is offered to a 3-year-old girl who has multiple painless skin swellings and a pathogenic mutation in *NF1* is found. Genetic testing of NF-1 gene in both parents shows no mutations. Which of the following genetic principles best describes this family?

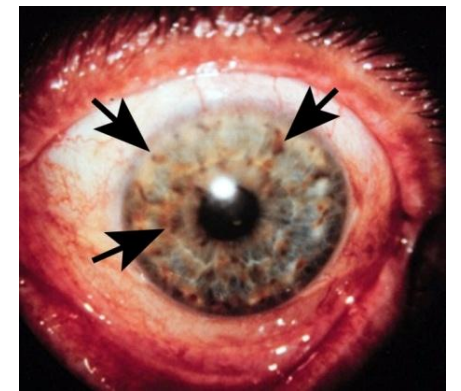
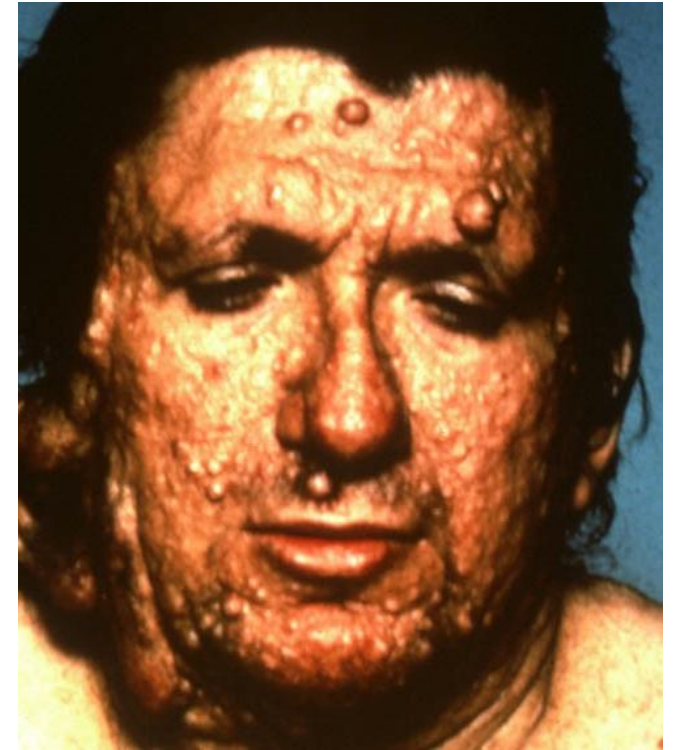


- | | |
|--------------------------|-----|
| A. De novo | 14% |
| B. Anticipated | 14% |
| C. Ex vivo | 14% |
| D. Germline mosaicism | 14% |
| E. Gain of function | 14% |
| F. Variable expressivity | 14% |
| G. Reduced penetrance | 14% |

In chapter 10 in the Korf textbook

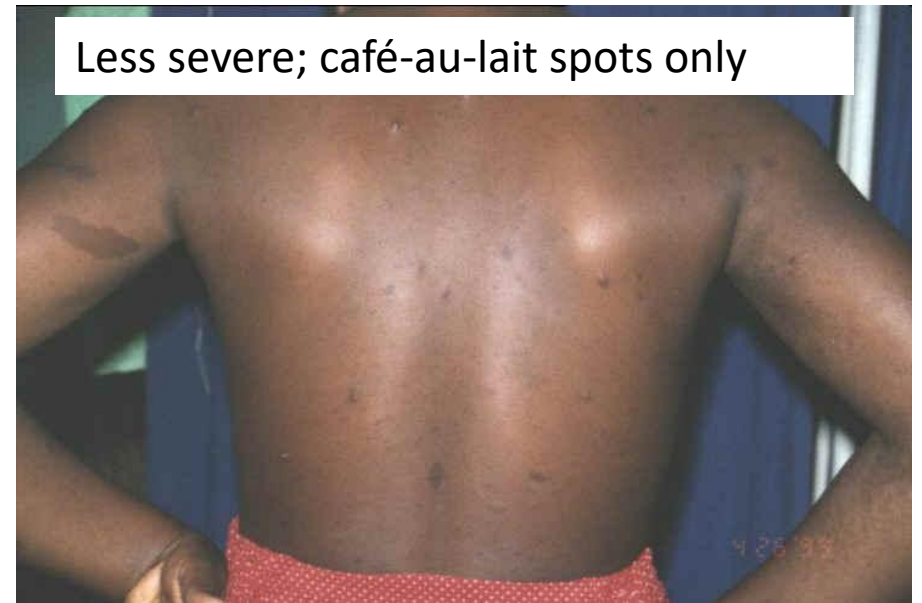
Neurofibromatosis (NF1)

- Mutations in *NF1* gene that codes for neurofibromin protein.
- NF1 may be caused by different mutations in the *NF1* gene (**allelic heterogeneity**).
 - Often, each person (or family) has a unique mutation
- *NF1* gene is very large, there are many places (*almost* infinite) where a **new mutation (de novo)** may inactivate the gene.
- Neurofibromin is a **tumor suppressor protein**
 - **Loss of function mutation**
 - More about *NF1* later in this course



Neurofibromatosis (NF1)

- Café-au-lait spots
- Neurofibromas: usually benign tumors; seen as swellings on the skin
- Lisch nodules in the iris of the eye
- NF1 shows **variable expressivity** but has **high penetrance**



Examination of a full-term newborn girl (IV-1) is shown. The family history is shown in the pedigree. Which of the following best explains the pathogenesis of this disorder in this patient?

A. Germline mosaicism

17%

B. Gain of function

17%

C. Homozygosity for the mutation

17%

D. Haploinsufficiency

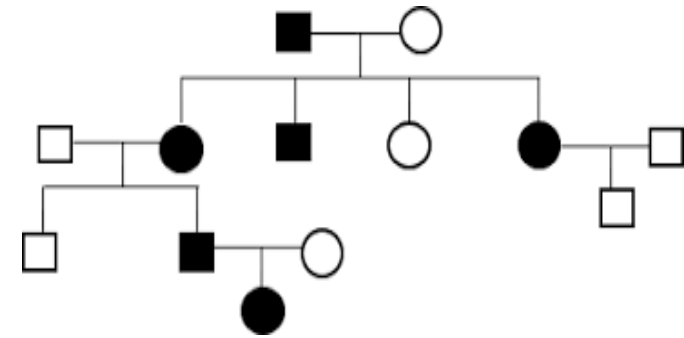
17%

E. Loss of function

17%

F. Locus heterogeneity

17%



Achondroplasia

- *FGFR3* mutations (Fibroblast growth factor receptor 3)
- *FGFR3* codes for a transmembrane receptor tyrosine kinase involved in signaling to control differentiation of cartilage to bone
- Mutations in *FGFR3* result in severe stunting of growth (evident at birth)
- **‘Gain of function’** mutation
 - Basically, the receptor signals when it is not supposed to.
- *FGFR3* gene has a **‘mutation hot spot’**; So **new mutations are likely to occur *de novo*** in index cases where there is no family history of the disorder



Achondroplasia

By Kyle Pacek (I took this photo of myself for my acting career) [GFDL (<http://www.gnu.org/copyleft/fdl.html>) or CC BY-SA 3.0 (<https://creativecommons.org/licenses/by-sa/3.0/>)]

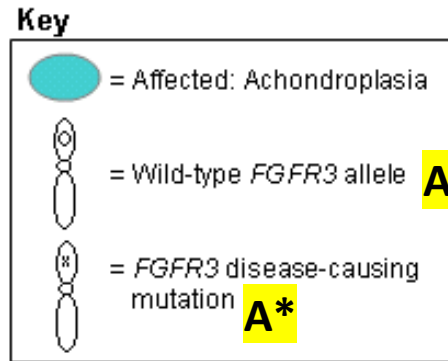
Read about achondroplasia on page 47 (Korf 4th edition)

Achondroplasia

- Students should understand that gain of function mutations are typically very specific and are likely to have very little (or no) allelic heterogeneity.



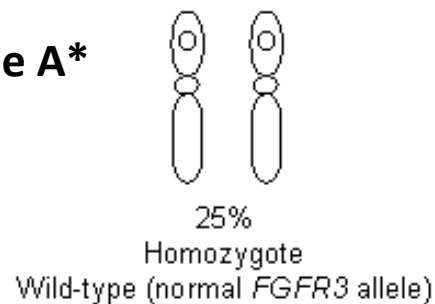
Recurrence risk when both parents are affected with achondroplasia



When both parents have achondroplasia dwarfism, Mendelian rules (*independent assortment of alleles*) explains why they may have a child with normal stature

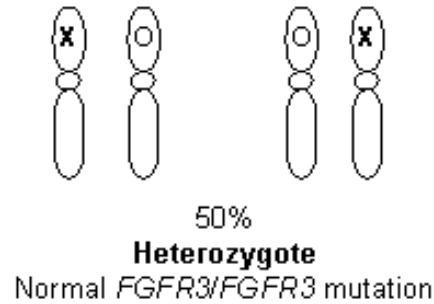
	Linda	
Richard	A	A*
A	AA	AA*
A*	AA*	AA*

A = functional allele
A* = *FGFR3* gain of function allele

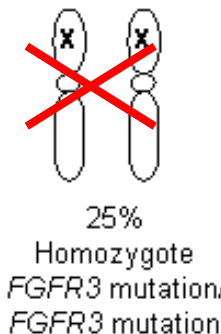


Phenotype: Unaffected

Possible Outcomes of Pregnancy



Phenotype: Affected with achondroplasia



Phenotype: Affected with lethal achondroplasia

Homozygosity for achondroplasia mutation is lethal during fetal development, or perinatal lethality. Hence, a person homozygous for the mutation cannot be observed (it is lethal)

Molecular explanation: why AD mutations manifest in the heterozygous state

- **Gain-of-function mutations:**
- Result from increased levels of gene expression or gene activity or development of a new function of the gene (*“attainment of a novel function”*)
 - Examples of Gain of Function include:
 - **Huntington disease** (*attainment of a novel function*)
 - **Achondroplasia** (*too much of the normal function*)
 - Many oncogene mutations are thought to be gain of function mutations (more on this later)

Homozygosity for alleles causing an AD disease: Prognosis always worse

- LDL Receptor defects leading to FH
 - Heterozygote (atherosclerosis in adulthood)
 - Homozygote (much more severe and childhood onset)
- Achondroplasia
 - Heterozygote has short stature
 - Homozygote has embryonic, fetal, or perinatal lethality
- Huntington disease
 - Homozygotes have earlier and more severe disease onset (Early childhood onset; rare)
- Homozygosity for an AD disorder is **extremely** rare
 - Unless **clearly** indicated, always assume a person with an AD disorder is a **heterozygote**

Autosomal Recessive

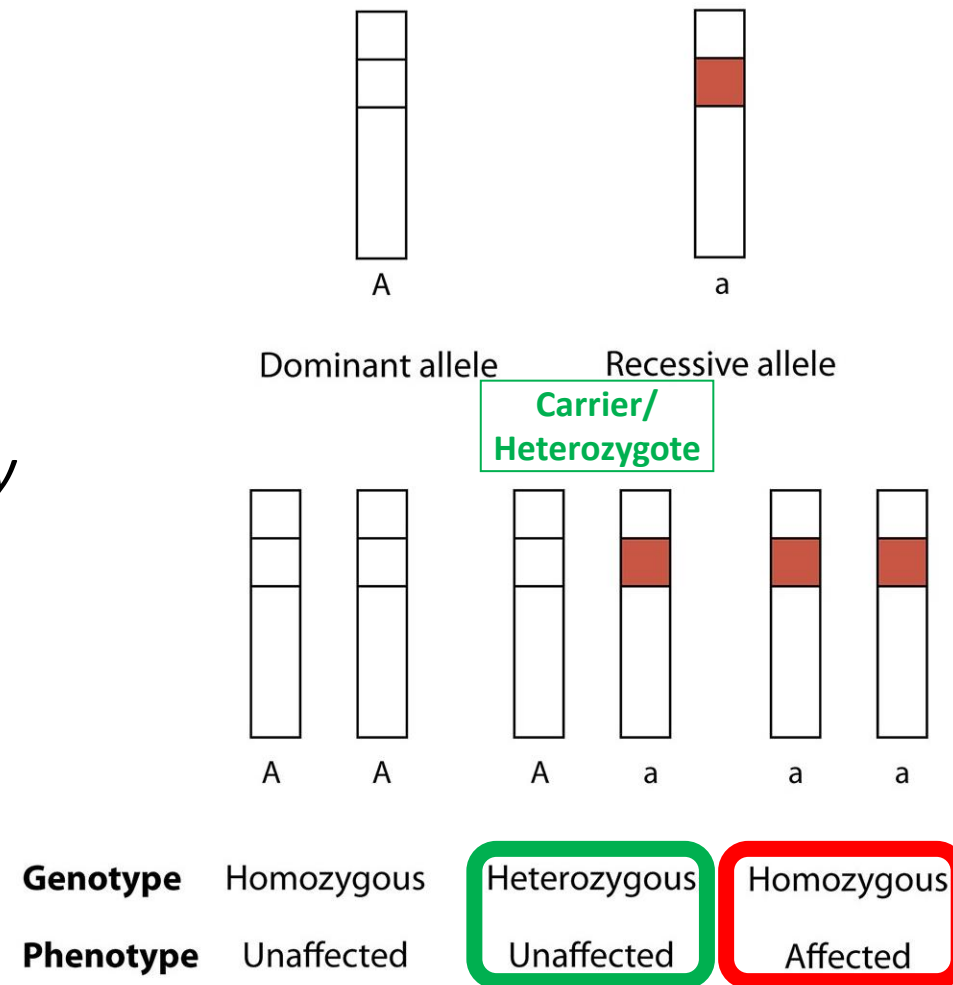
Autosomal Recessive disorders

- Cystic fibrosis
- Sickle cell anemia (sickle cell disease)
- Phenylketonuria (PKU)
- Tay-Sachs disease (Hexosaminidase A deficiency)
- Hemochromatosis (delayed age of onset)
- Homocystinuria
- Galactosemia
- Alpha 1-antitrypsin deficiency (AATD)
- SCID (Severe combined immunodeficiency) due to adenosine deaminase (ADA) deficiency
- Most enzyme deficiencies are AR
 - Beware those that are X-linked, and AIP (Acute intermittent porphyria is AD)

Gene name when discussed; Mode of inheritance, and main clinical features – each will be expanded upon as we progress through Medicine

Autosomal Recessive Disorders

- Expressed in the **homozygous state**
- Generally, both parents are **carriers** of a disease-causing mutant allele
- Carriers typically **phenotypically normal**
 - All humans carry recessive mutations
 - Vast majority of recessive alleles, *in any population*, are hidden in healthy heterozygote people



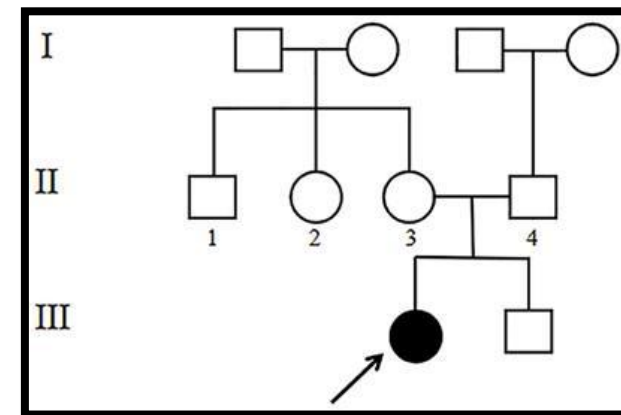
Human Genetics and Genomics, Fourth Edition. Bruce R. Korf and Mira B. Irons.
© 2013 John Wiley & Sons, Ltd. Published 2013 by John Wiley & Sons, Ltd.

Figure 3.2 Genotype and phenotype associated with a gene locus with dominant allele A and recessive allele a.

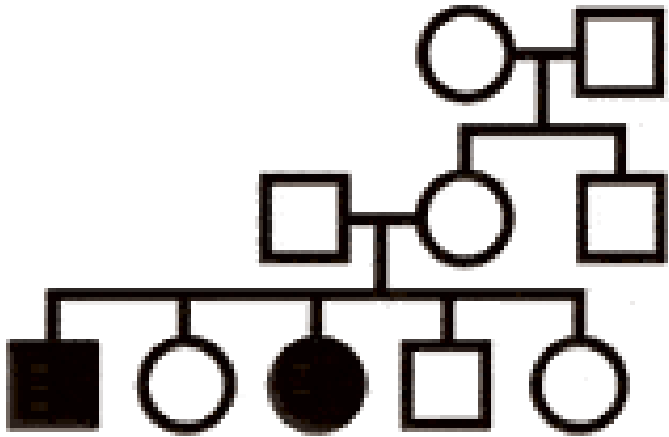
A 5-year-old girl (III-1) has skeletal abnormalities, developmental delay and learning problems. Laboratory studies show elevated homocysteine in serum and urine. She is diagnosed with homocystinuria. The parents are expecting another child. What is the best estimate that this next child will be similarly affected?

- | | |
|---------|----|
| A. 100% | 0% |
| B. 75% | 0% |
| C. 67% | 0% |
| D. 50% | 0% |
| E. 33% | 0% |
| F. 25% | 0% |
| G. 0% | 0% |

What is the risk that the parents are carriers (heterozygotes) of the mutation?

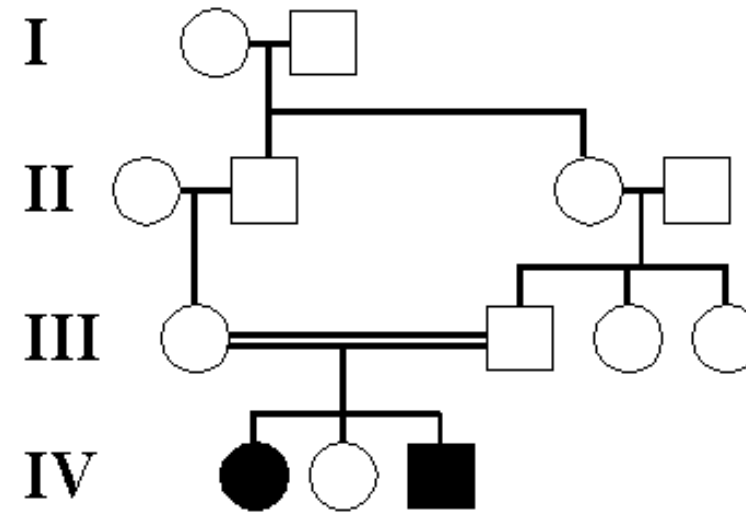


Autosomal Recessive disorders



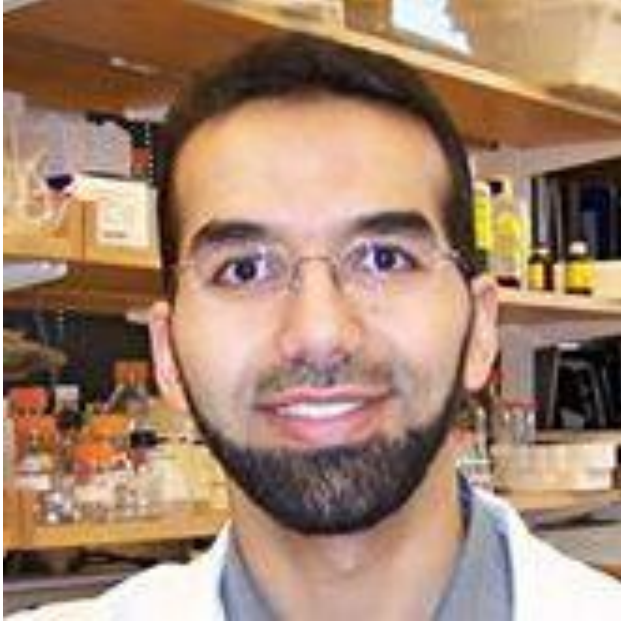
- Autosomal recessive disorders usually seen in only one generation of a pedigree (**horizontal inheritance**)
- Autosomal recessive disorders are expressed only in the homozygous state. Parents are usually **carriers** of the disease-causing allele. Siblings likely to be affected than others in the population (of course)
- Males and females are affected in almost equal frequency

Consanguinity *is sometimes* a finding in an autosomal recessive pedigree



- ✓ In many instances, consanguinity is usually **NOT** the cause of an AR disorder in a family
- ✓ However, the more rare a disorder is, the more likely consanguinity *might* play a role
- Related individuals may carry the same disease-causing allele
- It is not possible to trace the transmission of a mutation through the family without some kind of test (could be biochemical, or genetic) because *carriers are typically healthy and problems or phenotypic changes are often difficult to detect*
- ACOG provides recommendations for carrier screening of the more common genetic diseases including cystic fibrosis, spinal muscular atrophy, hemoglobinopathies (including sickle cell disease, α - and β -thalassemias), fragile X syndrome, and Tay-Sachs disease. ACOG recommends that carrier screening for cystic fibrosis and spinal muscular atrophy should be offered to all women considering pregnancy and all pregnant women, regardless of ethnicity, personal, or family history. In addition, a complete blood count together with red blood cell indices should be performed in all pregnant women and ideally to all women before pregnancy, to assess the risk of anemia as well as hemoglobinopathies. For other genetic diseases, ACOG recommends an ethnic group- and personal/family history-based approach.

Many governments are investing large sums of money to reduce first cousin marriages in their country



ASHG Honors Fowzan S. Alkuraya, MD, with the 2020 Curt Stern Award

<https://www.ashg.org/publications-news/press-releases/202007-curt-stern/>

<https://www.ajmc.com/view/genetic-carrier-screening-historial-perspective-and-overview>

Main point: Consanguinity increases the risk that a couple will have a child with a genetic disorder



Recurrence Risk for Autosomal Recessive Inheritance

A : Dominant normal allele of the gene

a : Recessive disease-causing allele of the gene

Both parents are usually carriers of the mutant allele

Normal Parent (Carrier) (Aa) X Normal Parent (Carrier) (Aa)

	Carrier mother	
Carrier father	A	a
A	AA	Aa
a	Aa	aa

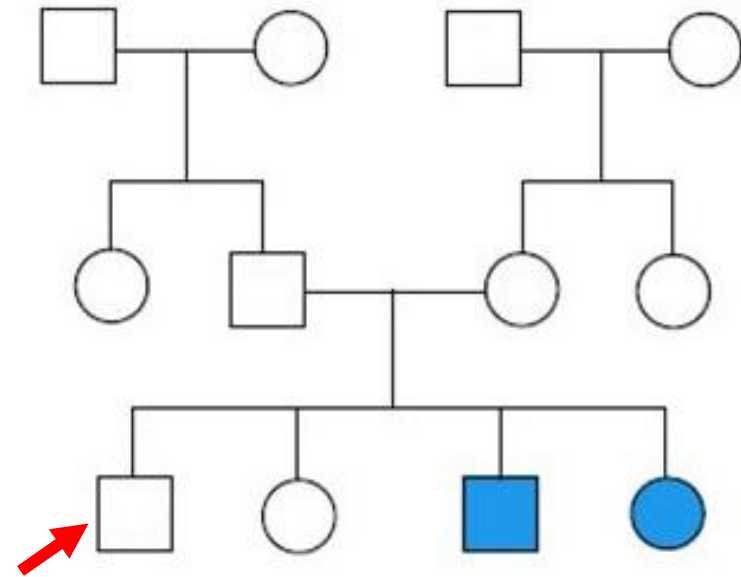
25%
homozygous
Normal

25%
homozygous
affected

75% Phenotypically normal children

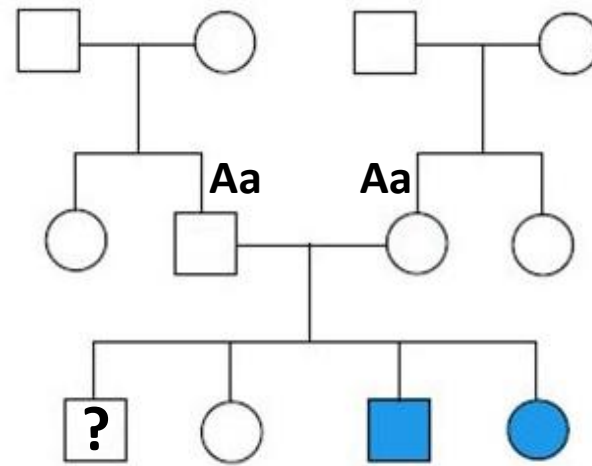
Individual III-1 (red arrow) has two younger siblings with galactosemia. Which of the following is the best estimate that he is a carrier of an allele that would be causative of this inherited disorder?

- | | |
|---------|----|
| A. 100% | 0% |
| B. 75% | 0% |
| C. 67% | 0% |
| D. 50% | 0% |
| E. 33% | 0% |
| F. 25% | 0% |
| G. 0% | 0% |



You must interpret from the vignette that the person under consideration is assumed to be healthy

Autosomal Recessive and the 2/3 rule



Caveat: This calculation only applicable to autosomal recessive disorders that are completely penetrant at an early age

II-2 and II-3 are carriers (heterozygotes)

What is the probability that III-1 is a carrier of the recessive allele?

III-1 **cannot** be homozygous for the mutant allele (aa) as he would manifest with the symptoms of the disorder

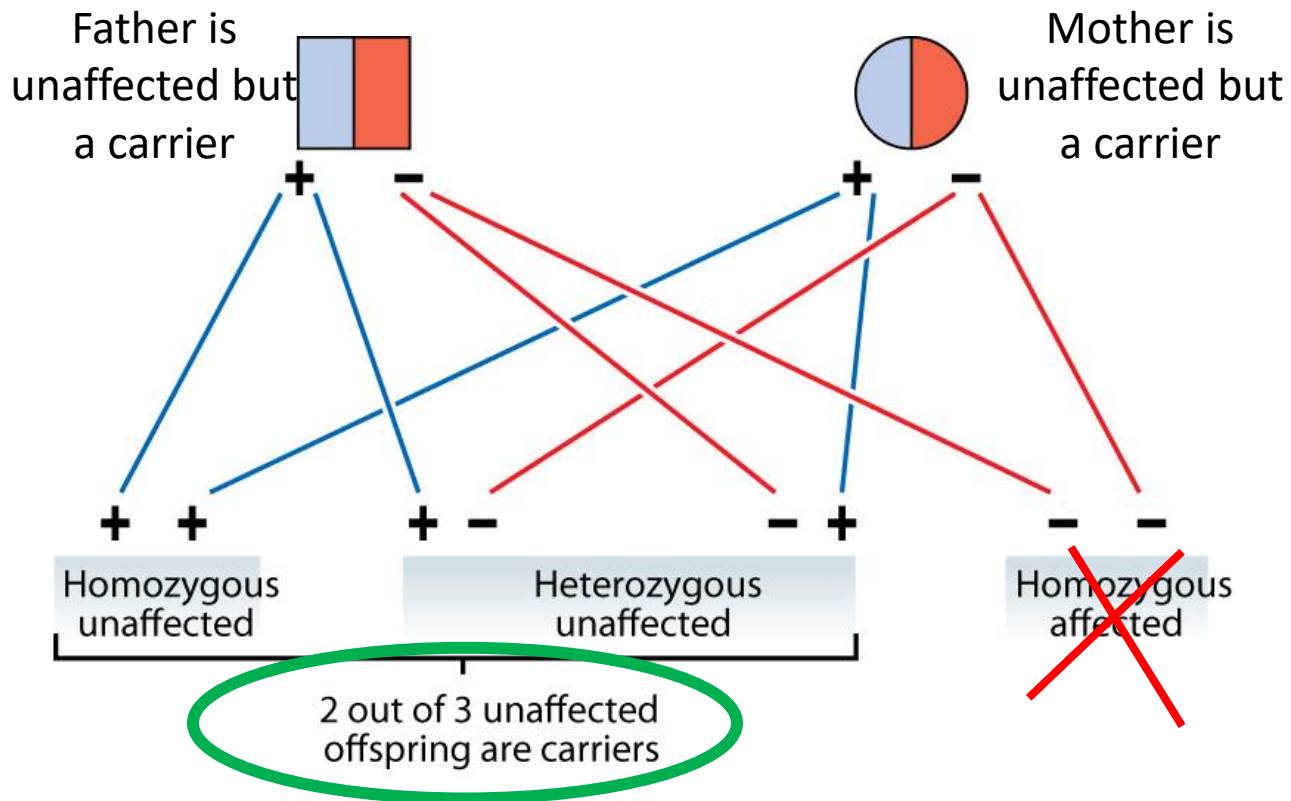
He could be

- Homozygous normal (AA)
- Heterozygous carrier (Aa)
- Heterozygous carrier (aA)

	Carrier mother	
Carrier father	A	a
A	AA	Aa
a	Aa	aa

He has a 2/3 chance of being a carrier (67%)

In Autosomal Recessive Sibships (brothers and sisters) what proportion of the **Non-Affected** Sibs would be carriers?



Explanation: if sib is not affected, that sib may not be 'aa' because he/she is not affected

Therefore, there are only three other genotypes available:

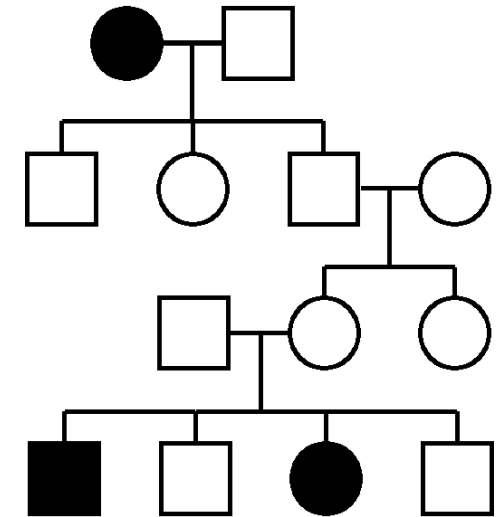
AA
Aa
aA

Notice that two (2) out of the three (3) are carriers:

So, for AR disorders, 2/3 of healthy sibs of a known affected person are carriers.

Caveat: disorder is highly penetrant and observable early in life.

Given is a pedigree showing some family members affected with sickle cell disease. Who are the obligate carriers in this family?



- A. I-2, II-1, II-3, III-1, IV-2
- B. II-1, II-2, II-3, III-1, III-2
- C. II-1, II-4, III-2, IV-2, IV-4
- D. II-2, II-4, III-1, III-2, III-3
- E. II-1, II-3, III-1, III-2, IV-4

0%

0%

0%

0%

0%

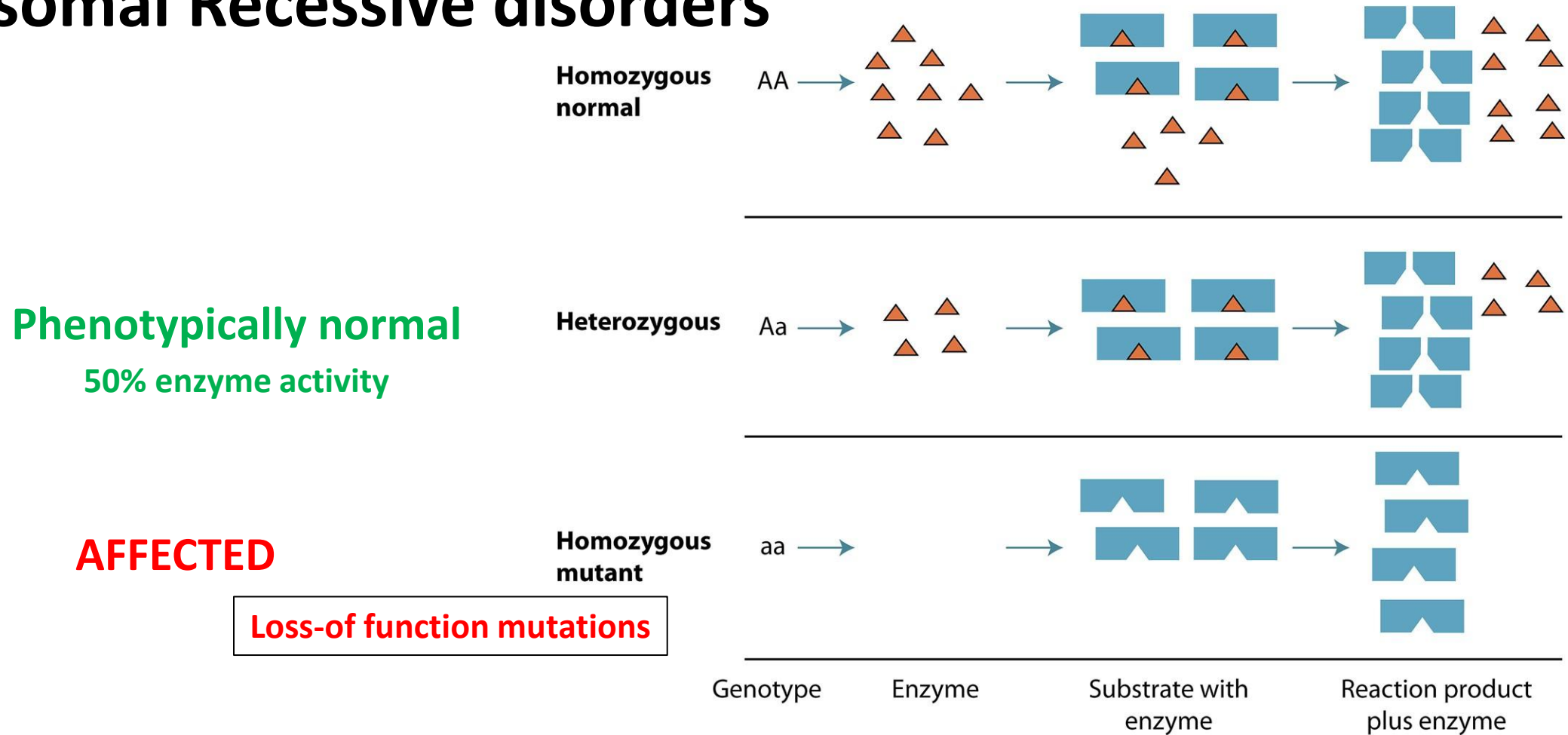


From GNET DLA

Molecular basis for autosomal recessive disorders

- **Most are loss-of function mutations:** result in either reduced activity (hypomorph) or complete loss of gene product (null allele); examples include enzyme deficiencies
- The protein produced from the normal allele (50%) in a carrier is generally sufficient to carry out the normal functions of the gene in the carrier state.
 - The carrier usually does **not have phenotypic manifestations**
 - A heterozygous carrier has 50% of the normal level of enzyme activity

Autosomal Recessive disorders



Human Genetics and Genomics, Fourth Edition. Bruce R. Korf and Mira B. Irons.
© 2013 John Wiley & Sons, Ltd. Published 2013 by John Wiley & Sons, Ltd.

Figure 3.7 Model explaining recessive transmission of most enzyme deficiencies. Normally, more than sufficient enzyme is synthesized to carry out a reaction. A heterozygote still has sufficient enzyme, but a homozygote for a mutation does not make enough enzyme to complete the reaction.

Tay-Sachs disease: example of an inborn error of metabolism

- Autosomal recessive loss of function of *HEXA* gene
- Deficient Enzyme is hexosaminidase A
- Lysosomal storage disorder
- Accumulation of a **glycolipid (Ganglioside) in the lysosome**
- Baby develops normally until mid-first year (6-8 months)
- Inexorable decline in neurological function

In born errors of metabolism (enzyme deficiencies) can manifest with many different phenotypes – see DLA for more examples

Compound heterozygosity

- Disorders caused by homozygous loss of function mutations (AR) often are caused by two *different* alleles in an individual

Father
CFTR / *CFTR*(del F508)

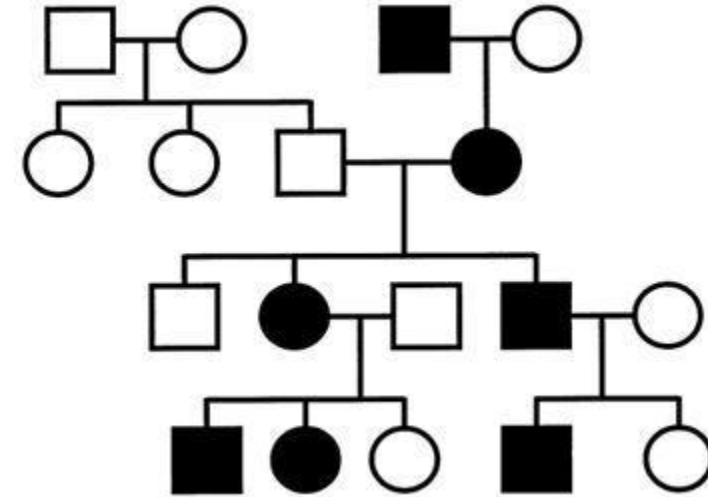
Both parents are healthy, but both
are carriers for pathogenic variants
in *CFTR* gene

Mother
CFTR / *CFTR*(R117H)

For this couple, each conception has 25% chance that the baby will have *CFTR* del F508/*CFTR* R117H and have CF

An affected baby from this couple would have CF as a result of compound heterozygosity

The affected people in this family have Charcot-Marie-Tooth disease (CMT). This disorder may exhibit many different inheritance patterns. Which mode of inheritance is followed in this specific family?



- A. Autosomal recessive
- B. Autosomal dominant
- C. X-linked dominant
- D. X-linked recessive
- E. Mitochondrial

0%

0%

0%

0%

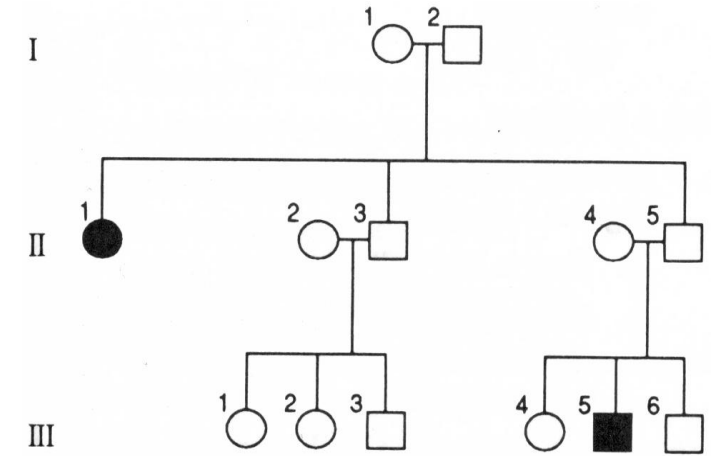
0%

From GNET DLA 13b

SOM.MKIII.BPM1.1.FTM.3.GNET.0027

See clinical snapshot 2.2 (p33) in Korf and Irons.
Do not memorize the gene names for this neurological disorder (there are many).

Individual II-1 has a disorder characterized by a relatively high carrier frequency in the population. The pedigree of the family is shown. Which of the following is the likely mode of inheritance?



- A. Autosomal recessive 0%
- B. Autosomal dominant 0%
- C. X-linked dominant 0%
- D. X-linked recessive 0%
- E. Mitochondrial 0%